The Hilbert–Schmidt basis-learning objective need not attain its spectral supremum

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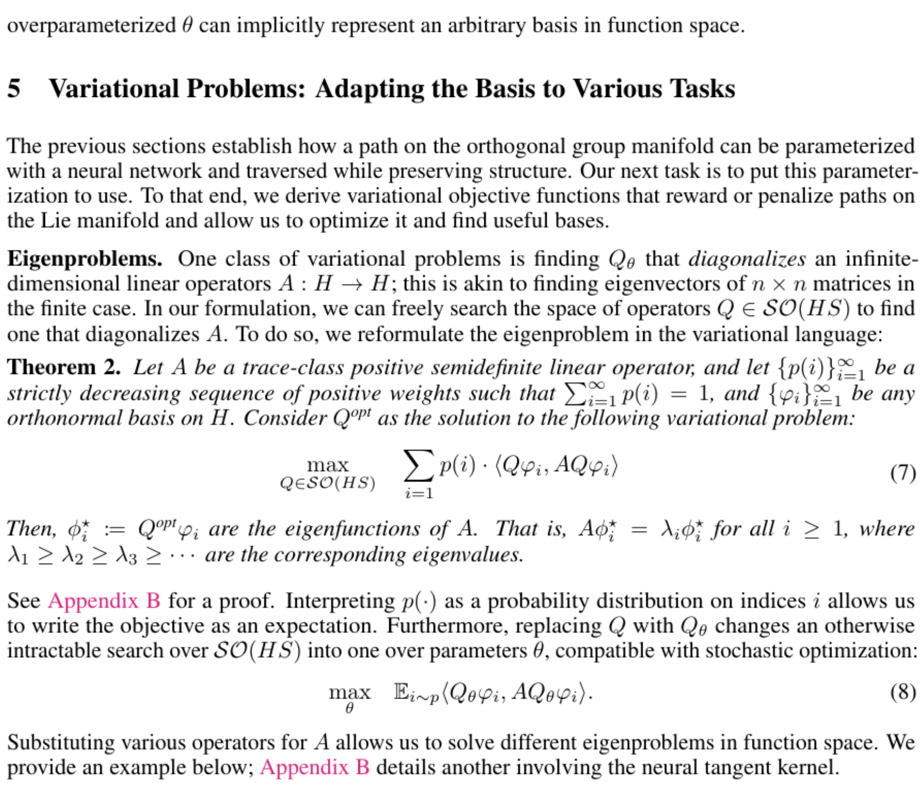
Status. Candidate full counterexample to optimizer existence, likely valid pending expert review.

1 Source target

Kamkari, Nabizadeh, and Solomon define $\mathcal{SO}(HS)$ using orthogonal flows generated by Hilbert–Schmidt skew-adjoint operators [1, Definition 1, PDF p. 4]. Their Theorem 2 treats the maximum

$$\max_{Q\in\mathcal{SO}(HS)} \sum_{i\geq 1} p(i)\langle Q\varphi_i, AQ\varphi_i\rangle \tag{1}$$

as having a solution Q^{opt} and concludes that the transformed basis is an eigenbasis ordered by decreasing eigenvalue [1, Theorem 2 and equation (7), PDF p. 5].



overparameterized θ can implicitly represent an arbitrary basis in function space.

5 Variational Problems: Adapting the Basis to Various Tasks

The previous sections establish how a path on the orthogonal group manifold can be parameterized with a neural network and traversed while preserving structure. Our next task is to put this parameterization to use. To that end, we derive variational objective functions that reward or penalize paths on the Lie manifold and allow us to optimize it and find useful bases.

Eigenproblems. One class of variational problems is finding Q_θ that *diagonalizes* an infinite-dimensional linear operators $A : H \to H$; this is akin to finding eigenvectors of $n \times n$ matrices in the finite case. In our formulation, we can freely search the space of operators $Q \in \mathcal{SO}(HS)$ to find one that diagonalizes A. To do so, we reformulate the eigenproblem in the variational language:

Theorem 2. *Let A be a trace-class positive semidefinite linear operator, and let $\{p(i)\}_{i=1}^{\infty}$ be a strictly decreasing sequence of positive weights such that $\sum_{i=1}^{\infty} p(i) = 1$, and $\{\varphi_i\}_{i=1}^{\infty}$ be any orthonormal basis on H. Consider Q^{opt} as the solution to the following variational problem:*

$$\max_{Q\in\mathcal{SO}(HS)} \quad \sum_{i=1} p(i)\cdot\langle Q\varphi_i, AQ\varphi_i\rangle \tag{7}$$

Then, $\phi_i^\star := Q^{opt}\varphi_i$ are the eigenfunctions of A. That is, $A\phi_i^\star = \lambda_i\phi_i^\star$ for all $i \geq 1$, where $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \cdots$ are the corresponding eigenvalues.

See Appendix B for a proof. Interpreting $p(\cdot)$ as a probability distribution on indices i allows us to write the objective as an expectation. Furthermore, replacing Q with Q_θ changes an otherwise intractable search over $\mathcal{SO}(HS)$ into one over parameters θ, compatible with stochastic optimization:

$$\max_{\theta} \quad \mathbb{E}_{i\sim p}\langle Q_\theta\varphi_i, AQ_\theta\varphi_i\rangle. \tag{8}$$

Substituting various operators for A allows us to solve different eigenproblems in function space. We provide an example below; Appendix B details another involving the neural tangent kernel.

The example below shows that the displayed supremum need not be attained. It lies entirely inside the paper's function-space/PCA setting: the domain is compact, the reference and target bases consist of continuous functions, and A is a positive trace-class covariance operator with continuous kernel.

2 Proof intuition

Pair the Fourier functions and rotate every pair through 90°. Give the resulting ordered basis the simple eigenvalues 2^{-i}, and use the same numbers as the objective weights. Exact spectral alignment requires moving *every* reference vector by a fixed distance. Such a change of basis is not identity plus Hilbert–Schmidt, hence is outside the source's admissible class. Rotating only the first m pairs is admissible and misses the optimum by exactly $1/(15 \cdot 16^m)$. Thus the objective has a maximizing sequence whose limit has escaped the Hilbert–Schmidt group.

3 Counterexample and exact value

Write $u \otimes v$ for the rank-one operator $f \mapsto \langle f, v\rangle u$.

Theorem 1. *Let $H = L^2([0,1];\mathbb{R})$ and let $(\varphi_i)_{i\geq 1}$ be an enumeration of the real Fourier orthonormal basis. There are strictly decreasing positive weights $p(i)$ summing to one and a positive trace-class operator A with a continuous integral kernel such that the objective in* (1) *satisfies*

$$\sup_{Q\in\mathcal{SO}(HS)} \sum_{i\geq 1} p(i)\langle Q\varphi_i, AQ\varphi_i\rangle = \frac{1}{3},$$

but no $Q \in \mathcal{SO}(HS)$ attains $1/3$. Moreover, admissible finite-rank exponentials Q_m attain exactly

$$\frac{1}{3} - \frac{1}{15\cdot 16^m} \qquad (m \geq 0).$$

The operator A is the covariance of a random continuous function.

Proof. Put $b_i = 2^{-i}$ and

$$B = \sum_{i\geq 1} b_i\, \varphi_i \otimes \varphi_i.$$

Define the orthogonal quarter-turn R blockwise by

$$R\varphi_{2k-1} = \varphi_{2k}, \qquad R\varphi_{2k} = -\varphi_{2k-1} \qquad (k \geq 1),$$

and set $A = RBR^*$. Then A is positive and trace class, with distinct eigenvalues b_i and corresponding ordered eigenvectors $e_i = R\varphi_i$.

The real Fourier functions are continuous and uniformly bounded by $\sqrt{2}$. Hence

$$a(x,y) = \sum_{i\geq 1} b_i e_i(x) e_i(y)$$

converges absolutely and uniformly on $[0,1]^2$, so it is a continuous integral kernel for A. If (ε_i) are independent Rademacher signs, then

$$X = \sum_{i\geq 1} b_i^{1/2} \varepsilon_i e_i$$

converges uniformly to a continuous random function because $\sum_i \sqrt{b_i} < \infty$, and $\mathbb{E}[X \otimes X] = A$.

Choose $p(i) = b_i = 2^{-i}$, so p is strictly decreasing and $\sum_i p(i) = 1$. For an orthogonal Q, set $S = R^*Q$. The objective is the Hilbert–Schmidt inner product

$$\begin{aligned} F(Q) &= \operatorname{tr}(BQ^*AQ) = \operatorname{tr}(BS^*BS) = \langle B, S^*BS\rangle_{\mathcal{S}_2} \\ &\le \|B\|_{\mathcal{S}_2}\, \|S^*BS\|_{\mathcal{S}_2} = \|B\|_{\mathcal{S}_2}^2 = \sum_{i\ge 1} 4^{-i} = \frac{1}{3}. \end{aligned}$$

Equality in Cauchy–Schwarz forces $S^*BS = B$. Thus $BS = SB$; since the eigenvalues of B are simple, there are signs $\epsilon_i \in \{-1, 1\}$ such that $S\varphi_i = \epsilon_i\varphi_i$. Every equality case therefore has

$$Q\varphi_i = \epsilon_i R\varphi_i, \qquad \|Q\varphi_i - \varphi_i\|^2 = 2 \quad (i \ge 1).$$

Consequently $Q - I$ is not Hilbert–Schmidt.

On the other hand, every member of the source's $\mathcal{SO}(HS)$ satisfies $Q - I \in \mathcal{S}_2$. Indeed, for $Q = e^K$ with Hilbert–Schmidt skew-adjoint K,

$$e^K - I = \int_0^1 e^{tK} K \, dt \in \mathcal{S}_2.$$

(The same conclusion follows from Duhamel's formula for the broader time-dependent flows used later in the source.) Hence no admissible Q can be an equality case.

It remains to prove that $1/3$ is nevertheless the supremum. For $m \ge 0$ let

$$K_m = \frac{\pi}{2} \sum_{k=1}^{m} \bigl(\varphi_{2k} \otimes \varphi_{2k-1} - \varphi_{2k-1} \otimes \varphi_{2k}\bigr), \qquad Q_m = e^{K_m}.$$

Then K_m is finite-rank, Hilbert–Schmidt, and skew-adjoint; Q_m agrees with R on the first m pairs and is the identity on the remaining pairs. Thus $Q_m \in \mathcal{SO}(HS)$. On a tail pair, the two eigenvalues are swapped, and direct summation gives

$$\begin{aligned} F(Q_m) &= \sum_{i=1}^{2m} 4^{-i} + \sum_{k>m} \bigl(2^{-(2k-1)}2^{-2k} + 2^{-2k}2^{-(2k-1)}\bigr) \\ &= \frac{1}{3} - \sum_{k>m} 16^{-k} = \frac{1}{3} - \frac{1}{15 \cdot 16^m}. \end{aligned}$$

These values increase to $1/3$, proving both the supremum formula and its nonattainment. □

4 Structural diagnosis and scope

The obstruction is topological, not spectral. The weighted objective is insensitive to changes far out in the basis, whereas the Hilbert–Schmidt group requires the squared displacements of *all* basis vectors to be summable. The sequence Q_m converges strongly to R, but $R - I \notin \mathcal{S}_2$; it does not converge to R in Hilbert–Schmidt topology. This is precisely the escape mechanism that destroys compactness of the optimization problem.

The result supplies a full negative answer to the implicit optimizer- existence question in Theorem 2 and to the surrounding claim that one can freely search the stated class to find a diagonalizer for every admissible A. It does *not* refute the conditional assertion that, when a maximizer exists, strict weights order its eigenvectors. Nor does it rule out optimization over a larger orthogonal class or under an additional compatibility assumption ensuring that an ordered eigenbasis differs from the reference basis by a Hilbert–Schmidt operator.

The source appendix also estimates a Hilbert–Schmidt norm of an orthogonal operator itself. In infinite dimension such an operator is never Hilbert–Schmidt; only differences such as $Q - I$ may be. The argument above avoids that issue and uses only the ideal property of $\mathcal{S}_2$.

5 Verification and novelty

The accompanying exact-arithmetic regression script checks the finite-pair objective, the closed formula for the defect, and the divergence of the required squared basis displacement. It is not needed for the proof.

A bounded search on 27 August 2026 covered all four run indexes, the current arXiv record, the exact paper title and theorem name, and searches combining Hilbert–Schmidt orthogonal groups, diagonalization, optimizer existence, and nonattainment. The source remains at version 1, and no correction or later response was found. Novelty confidence is moderate pending specialist review.

Human review recommendation. Check that the source intends $\mathcal{SO}(HS)$ literally as Definition 1 rather than the full orthogonal group. Under either the stationary definition or the later time-dependent Hilbert–Schmidt-flow definition, the necessary condition $Q - I \in \mathcal{S}_2$ used above is immediate.

References

[1] H. Kamkari, M. S. Nabizadeh, and J. Solomon, *Learning Orthonormal Bases for Function Spaces*, arXiv:2605.19959v1 (2026).